

Minsung Kim

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Education

University of Toronto - Computer Science Specialist

GPA 3.7/4 | *University of Toronto International Scholar Award*

September 2025 – June 2029

Skills

Languages: C/C++, C#, Python, Java, TypeScript, JavaScript, SQL, HTML5/CSS, PostgreSQL, MySQL, C/C++, C#, Bash, R

Tools/Frameworks: React, Next.js, Vite, Zustand, Tailwind CSS, Material UI, jQuery, Express.js, RESTful APIs, WebSockets, PostgreSQL, MySQL, Supabase, Python, Scikit-learn, K-Means, Open CV, AWS, Git, GitHub, Linux, Postman, Bcrypt

Project

UofT Student Life Tracker

July 2026 – August 2026

- Developed a student dashboard integrating UofT resources using React, Express, PostgreSQL, and Material UI to centralize the navigation between separate platforms
- Built 13 RESTful API endpoints for a calendar system that manages events, tasks, and deadlines, enabling the users to organize their coursework in a single interface
- Secured user authentication by implementing server-side Express sessions, hardened connect.sid cookies, and bcrypt password hashing

Cluster Grade AI

March 2026

Hack the Student Life

AWS Services & University of Toronto

- Developed Cluster Grade AI with a team of three, an AI grading platform that extracts text from scanned handwritten assignments using AWS, Python Scikit-Learn, and Sentence Transformers to automate analysis of student submissions
- Built an OCR-to-clustering pipeline that converts scanned assignments into semantic embeddings and applies K-Means to group recurring logic errors, achieving a 0.72 silhouette score across 10 test submissions
- Integrated Amazon Bedrock with instructor rubrics to generate rubric-aligned deductions and feedback, allowing one TA review to serve entire groups instead of rewriting repetitive feedback
- Designed a human-in-the-loop grading workflow requiring TA approval before applying AI-generated feedback, targeting a 40–60% reduction in grading time for large introductory courses

WatchLine

March 2026

Hack Canada

SPUR Innovation

- Collaborated in a team of three to create WatchLine, a real-time AI transit safety system using YOLOv8-Pose, Python, and open CV to track human pose and classify falls, aggression, crouching, lying down, and erratic movement from live camera streams
- Built an asynchronous FastAPI and WebSocket backend that processes live video, maintains per-person motion histories, automatically records critical incidents as video clips, and broadcasts detection events to connected clients in real time
- Integrated Google Gemini 2.0 Flash to convert video evidence of detected incidents into English dispatch summaries, using Supabase/PostgreSQL to store incident data
- Developed a monitoring dashboard with React, Vite, and Zustand that renders MJPEG camera streams, displays real-time safety alerts, and provides incident clip playback for rapid event review

Experience

ResNet Technical Assistant

May 2026 – Present

University of Toronto

- Troubleshoot Wi-Fi, Ethernet, and eduroam connectivity issues across Chestnut, Oak House, and Graduate House, resolving network access problems for residential users
- Assist residents with wired and wireless device connectivity, including Ethernet cables, USB-A/USB-C adapters, HDMI cables, and gaming consoles requiring MAC address registration
- Monitor residential network infrastructure through regular access-point and connectivity checks, while restoring disconnected Ethernet and university phone lines through troubleshooting and power cycling
- Troubleshoot eduroam authentication issues on Android devices, configuring CA certificates and network security settings to restore connectivity, while providing guest-network alternatives when device configurations prevented authentication.